

The Universal Research Companion

Use Cases, Features & Lifecycle Gap Analysis

Research Companion project - strategy & gap analysis (June 2026 deep-research pass)

*An evidence-based assessment of the present tool — the **Research Companion** knowledge-graph foundation, the shipped deterministic reference validator, and the spec'd novelty engine — against the full research-paper lifecycle, with the missing use cases that would make it genuinely universal.*

Basis: Deep-research pass — 5 angles, 26 sources, 124 claims extracted, 25 adversarially verified (**18 confirmed, 7 refuted**).

Scope: All disciplines (CS/AI, STEM, biomedical, humanities, social sciences).

Status legend: **[BUILT]** BUILT · **[SPEC'D]** SPEC'D · **[PLANNED]** PLANNED · **[MISSING]** MISSING

1 — Executive summary

The single most valuable thing this tool can be is the thing nobody has built: **one product that spans the entire research lifecycle on a persistent knowledge graph**. Today's AI research tools are *stage-siloed* — strong at one slice, absent everywhere else. A comprehensive 2025 survey (arXiv:2503.01424) names "*end-to-end integration across lifecycle stages*" as a key open challenge, and even the most ambitious agent systems (AI Scientist v2, InternAgent, Agent Laboratory) remain incomplete. Research Companion's cross-paper graph is the right backbone for that integration.

What the evidence says to do:

- **Lead with the two underserved, low-competition use cases:** rebuttal / reviewer-response assistance (almost no general tool does it) and deep reproducibility / artifact checking (86.7% of studied papers had reproducibility "smells").
- **Keep the deterministic citation validator front-and-centre** — it already does the one thing the crowded compliance tools do *not*: live CrossRef/OpenAlex lookups that catch **fabricated** citations. 51% of students fear AI-hallucinated references, so this is also the trust anchor.
- **Treat compliance, plagiarism, and venue-matching as table-stakes**, not differentiators — they are mature, crowded categories (Penelope.ai, CheckMyManuscript, iThenticate, Researcher.Life).
- **Build statistical soundness as a bundle of narrow deterministic detectors** (Statcheck, GRIM), not a general LLM check.
- **Win adoption through trust + integration:** explainable "no black box" output, and first-class Zotero / LaTeX-Overleaf / Word interoperability.

2 — Critical analysis of the present condition

Graded honestly, the tool today is a **strong foundation with one shipped feature and a clear, evidence-backed plan** — but it currently touches only a narrow band of the lifecycle (prior-art mapping + citation integrity). The novelty engine is designed but unbuilt, and the entire pre-writing / ideation stage — which the research flags as high-value — is absent.

Component	State	Honest assessment
Research Companion knowledge graph (ingest, extract, cluster, chat, viz)	[BUILT] BUILT	The real strength & the moat. Cross-paper concept graph + community detection is exactly the integration backbone the field lacks. Underused so far — it only powers search/chat, not yet novelty or gap analysis.
Deterministic reference validator (CrossRef/OpenAlex, fabrication detection)	[BUILT] BUILT	Shipped this cycle, 46 tests, TDD. Does the one thing crowded compliance tools do NOT: live-lookup fabrication detection. Directly answers the #1 user fear. Needs more retrievers (S2/arXiv/DBLP) + reference-parsing hardening.
Novelty engine (contribution extraction, prior-art comparison, evidence verify)	[SPEC'D] SPEC'D	Designed in detail, not built. The intended flagship differentiator. Validated by OpenNovelty/GraphMind, but until built it is a promise, not a capability.
Reviewer critique · venue fit · framing advisor	[PLANNED] PLANNED	On the roadmap only. Reviewer critique & venue fit are already shipping elsewhere (Paperpal, Researcher.Life) — must differentiate on depth/evidence-linkage, not existence.
Ideation · writing-quality · reproducibility · rebuttal · stats · integrations	[MISSING] MISSING	Not in scope yet. This is where the biggest universal-coverage gaps — and the least-contested opportunities (rebuttal, reproducibility) — live.

Verdict: a credible foundation aimed at the right wedge (citation integrity + prior-art), but currently a **single-stage** tool wearing a **universal** ambition. Closing that distance is the roadmap below.

3 — The full lifecycle: use-case coverage map

Every use case a universal research tool should cover, the strongest existing players, and where this tool stands. This is the master feature inventory.

Stage A · Pre-writing / Ideation — *largely an open field*

Use case	Who does it now	Status	Priority
Research-gap analysis	Few credible tools; LLMs ad-hoc	[MISSING] MISSING	HIGH — graph-native fit
Idea / hypothesis generation	AI Scientist, Elicit (partial)	[MISSING] MISSING	Medium
Research-question formulation	Generic LLMs	[MISSING] MISSING	Medium
Literature-review automation	Elicit, Undermind, SciSpace	[MISSING] MISSING	High (crowded)
Method / dataset finding	PapersWithCode, Google Dataset	[MISSING] MISSING	Medium
"Has this been done?" / feasibility	Connected Papers (weakly)	[SPEC'D] SPEC'D	HIGH — novelty engine

Why it matters: ideation is where AI adoption is already near-universal among students, yet credible dedicated tools are thin. Research-gap analysis and "has this been done?" map directly onto the persistent knowledge graph — this is Research Companion's most natural extension and feeds straight into the novelty engine.

Stage B · Writing / Structuring

Use case	Who does it now	Status	Priority
Clarity / readability	Paperpal, Writefull, Grammarly	[MISSING] MISSING	Table-stakes
Argument / logical-flow check	Paperpal AI Review (shallow)	[PLANNED] PLANNED	High differentiator
Statistical soundness	Statcheck, GRIM (narrow)	[MISSING] MISSING	HIGH — bundle detectors
Methodology soundness	Fragmented / manual	[PLANNED] PLANNED	High
Claim–evidence linkage	Scite (citation stance)	[SPEC'D] SPEC'D	HIGH — graph + novelty
Related-work completeness	Manual; partial in LR tools	[SPEC'D] SPEC'D	HIGH — graph-native
Figure / table quality	Minimal tooling	[MISSING] MISSING	Low/Medium

Why it matters: statistical soundness is high-value but must be a **bundle of narrow deterministic detectors** — Statcheck (96–99.9% accuracy, recalculates reported p-values) and GRIM (>92%, mean-plausibility) are individually excellent but narrow; no single tool covers statistical review completely. Claim–evidence linkage and related-work completeness are graph-native and reuse the novelty machinery.

Stage C · Submission / Post-submission

Use case	Who does it now	Status	Priority
Citation / reference integrity	Compliance tools (format only)	[BUILT] BUILT	SHIPPED — keep central
Venue / journal fit	Researcher.Life (43k journals)	[PLANNED] PLANNED	Medium (crowded)
Reviewer-style critique	Paperpal AI Review	[PLANNED] PLANNED	Differentiate on depth
Rebuttal / reviewer-response	Almost nobody	[MISSING] MISSING	HIGHEST — open field
Reproducibility / artifact check	Niche; 86.7% papers fail	[MISSING] MISSING	HIGH — underserved
Ethics / compliance declarations	Penelope.ai, CheckMyManuscript	[PLANNED] PLANNED	Table-stakes
Plagiarism / self-plagiarism	iThenticate (1,500+ pubs)	[MISSING] MISSING	Table-stakes infra
AI-content disclosure	Emerging requirement	[MISSING] MISSING	Medium

Why it matters: rebuttal automation is repeatedly called "*largely underexplored*" in 2025–26 research yet is a crucial peer-review step — and essentially no general-purpose researcher tool offers it. Reproducibility checking has concrete, machine-checkable gaps. These two are the clearest open fields.

4 — Where to compete: crowded vs. open

[BUILT] OPEN FIELD — attack here

- **Rebuttal / reviewer-response** — "largely underexplored" in 2025–26; near-zero general tools.
- **Reproducibility / artifact checking** — concrete machine-checkable gaps (missing prompt templates, inference parameters).
- **Research-gap analysis** — perfect fit for the persistent knowledge graph.
- **Fabricated-citation detection** — already shipped; competitors only do format checks.
- **Cross-stage memory** — one graph carrying context from idea to rebuttal.

[MISSING] CROWDED — table-stakes only

- **Pre-submission compliance** — Penelope.ai (30+ checks), CheckMyManuscript (80+).
- **Plagiarism screening** — iThenticate via CrossRef Similarity Check (1,500+ publishers).
- **Venue matching** — Researcher.Life, 43k journals.
- **Section-by-section AI critique** — Paperpal "AI Review" coach.
- **Language / clarity polish** — Writefull, Grammarly, Paperpal.

5 — Recommended roadmap additions (beyond current plan)

New use cases the research says to add, ordered by opportunity × fit with the existing graph:

#	New capability	Why now
1	Rebuttal assistant — draft point-by-point reviewer responses grounded in the paper + graph	Near-empty market; high pain; reuses contribution + evidence-verification machinery.
2	Reproducibility / artifact linter — discipline-aware checklist (code/data links, prompts, params, protocols)	86.7% of papers fail; deterministic and machine-checkable; strong trust signal.
3	Research-gap & "has this been done" analysis on the knowledge graph	Opens the entire ideation stage; uniquely suited to Research Companion; feeds the novelty engine.

#	New capability	Why now
4	Statistical-soundness bundle — Statcheck + GRIM-style deterministic detectors	High value, proven accuracy; must be narrow detectors, not a general LLM.
5	Integrations layer — Zotero, LaTeX/Overleaf, Word import/export	Integration friction is a documented adoption killer; unlocks real workflows.
6	Explainability everywhere — every flag shows its evidence, no black box	Trust is the decisive adoption driver; 51% fear AI fabrication.

6 — Competitor teardown (appendix)

How the named players cover the lifecycle, and what a universal tool must beat. Grounded in the verified findings plus the earlier source-level teardown (COMPETITOR_ARCHITECTURES.md).

Tool	What it does well	Limits	What we must beat
Penelope.ai	30+ transparent, explainable compliance checks: headings, abstract subheadings, title-page elements, citation style (Harvard/Vancouver), citation-reference matching, all five ethics declarations, word/figure/table counts. Journal-integrated, white-labeled per journal.	Compliance/format only — explicitly no novelty, no scientific-merit critique . Citation checks are format/consistency, not live-lookup.	Match its explainability ("no black box") while adding the scientific judgment it deliberately omits.

Tool	What it does well	Limits	What we must beat
CheckMyManuscript	80+ checks across structure, citations, metadata, language quality, formatting consistency: missing entries, mismatches, ordering, duplicate merging, DOI/URL format validation, self-citation analysis.	DOI/URL check is format/consistency , not a CrossRef/OpenAlex live lookup. No fabricated-citation detection.	Our shipped validator already beats it on fabrication detection via live lookups.
Researcher.Life (Journal Finder)	Matches manuscript/abstract/topic against 43,000+ journals (CrossRef/DOAJ/Scopus/WoS); four input methods; scope-fit "Degree of Match" + similar papers.	Venue-matching only; (impact-factor/acceptance-rate coverage uncertain — a claim that it omits them was <i>refuted</i> , so re-verify).	Richer venue <i>strategy</i> (fit + acceptance odds + fit-to-contribution), not just scope match.
Paperpal (AI Review)	Section-by-section AI critique as a "virtual research coach"; language/clarity polish; broad integrations.	Critique is relatively shallow ; not evidence-linked to prior art or the knowledge graph.	Differentiate on depth + evidence-linkage (every critique tied to graph/prior-art), not mere existence.
iThenticate	Industry-standard similarity screening against 1,500+ publishers via CrossRef Similarity Check; deep publisher-workflow integration.	Similarity only; no scientific evaluation. Mature infrastructure, not a differentiator.	Don't rebuild — integrate or treat as table-stakes.

Tool	What it does well	Limits	What we must beat
OpenNovelty (research)	Verifiable novelty assessment: 4-phase pipeline (extract claims → retrieve prior work → contribution-level comparison → cited report) with evidence verification. Deployed on 500+ ICLR 2026 submissions.	CS/AI-only; single-purpose; not integrated with venue/compliance linting.	Match its verifiable, citation-grounded output; exceed it on lifecycle breadth + multi-discipline.
GraphMind (research)	Open-source hybrid LLM + retrieval + knowledge-graph view for novelty; arXiv/Semantic Scholar integration; citation-context polarity typing.	CS/AI-only; ~60% of its code is experiment scaffolding, not product.	The closest analog to Research Companion — beat it on product polish, breadth, and ease of use.

7 — Caveats & what we could NOT confirm

- **Vendor self-reports:** "30+/80+ checks", "43,000 journals", "1,500+ publishers" are marketing figures, not independently audited — directional, not exact.
- **Discipline skew:** the strongest reproducibility evidence (arXiv:2512.00651) is CS/LLM-specific (prompt templates, inference params don't exist in humanities/wet-lab). A truly universal tool needs discipline-specific checklists — real R&D, not a config flag.
- **Demand data is UK-undergraduate-centric** (HEPI 2025). What PIs and non-CS researchers actually ask for remains an open question — validate before over-fitting to student needs.
- **7 claims were refuted** and excluded — e.g. that Paperpal lacks Overleaf/LaTeX or journal-fit features; that Researcher.Life omits impact-factor/acceptance-rate metrics; that general LLMs are dramatically worse (~52%) at stat-error detection; and a proposed rigid 4-stage lifecycle model. Competitor feature boundaries need direct re-verification before being used to claim differentiation.
- **Ideation-tool comparisons did not survive verification** — a notable evidence gap, given ideation is a priority stage. Treat Stage-A competitor claims as unconfirmed.

8 — Key sources

- AI-for-Research survey — arXiv:2503.01424 (end-to-end integration gap)
- Rebuttal generation (DRPG) — arXiv:2601.18081; corroborated by Defend, RbtAct, Re2, ReviewMT
- LLM-for-SE reproducibility study — arXiv:2512.00651 (640 papers, 86.7% with smells)

- Penelope.ai & CheckMyManuscript (compliance checks); Researcher.Life Journal Finder
- Statcheck / GRIM validation review — PMC12722777; iThenticate / CrossRef Similarity Check
- HEPI Student Generative AI Survey 2025 (51% fear hallucinated citations); Paperpal AI Review; Overleaf–Zotero docs

Generated from a verified deep-research pass — 18 confirmed findings of 25 adversarially tested. Claims carry source citations; refuted claims were excluded. This document maps the product; it does not itself constitute peer-reviewed evidence.